

Ensemble forecasting for product futures prices using variational mode decomposition and artificial neural networks

[Author & abstract](#)
[Download](#)
[21 References](#)
[16 Citations](#)
[Most related](#)
[Related works & more](#)
[Corrections](#)

Author

Listed:

[Registered:](#)

- Liu, Weiping
- Wang, Chengzhu
- Li, Yonggang
- Liu, Yishun
- Huang, Keke

Abstract

Product futures are materials support of industrial and society, and forecasting the product futures prices is of great significance to society and enterprises. However, the product futures prices sequences often show non-stationary and non-linear characteristics, so it is significant yet challenging to forecast product futures prices accurately. To cope with this issue, this paper proposes a novel approach that combines variational mode decomposition (VMD) and artificial neural network (ANN) into a “decomposition and ensemble” framework, and the so-called VMD-ANN method is presented for futures prices forecasting. In particular, the proposed approach innovatively introduces the VMD method to transform the problem of futures prices forecasting with high volatility into multiple component time series forecasting with unique central frequency, which greatly reduces the difficulty for forecasting futures prices. Then, the ANN method is utilized to forecast all components independently. Finally, the prediction results of each component are integrated into the final prediction results. In order to demonstrate the performance of the proposed method, four benchmark futures prices, the West Texas Intermediate (WTI) crude oil prices, London Metal Exchange (LME) zinc prices, the New York Mercantile Exchange(NYMEX) natural gas prices and the Commodities Exchange(COMEX) gold prices, are introduced to show the superiority of the proposed method. The experimental results show that whether single-step-ahead forecasting or multi-step-ahead forecasting, the numerical accuracy and trend accuracy of the proposed VMD-ANN significantly outperform some state-of-the-arts methods on both energy futures and metal futures, which verify that the proposed VMD-ANN method can effectively forecast non-stationary and non-linear futures prices series.

Suggested Citation

↓ Liu, Weiping & Wang, Chengzhu & Li, Yonggang & Liu, Yishun & Huang, Keke, 2021. "[Ensemble forecasting for product futures prices using variational mode decomposition and artificial neural networks](#)," [Chaos, Solitons & Fractals](#), Elsevier, vol. 146(C).

Handle: *RePEc:eee:chsofr:v:146:y:2021:i:c:s0960077921001740*

DOI: 10.1016/j.chaos.2021.110822

Export reference



as

HTML



IDEAS is a [RePEc](#) service. RePEc uses bibliographic data supplied by the respective publishers.